Next week Tuesday : Quiz 2

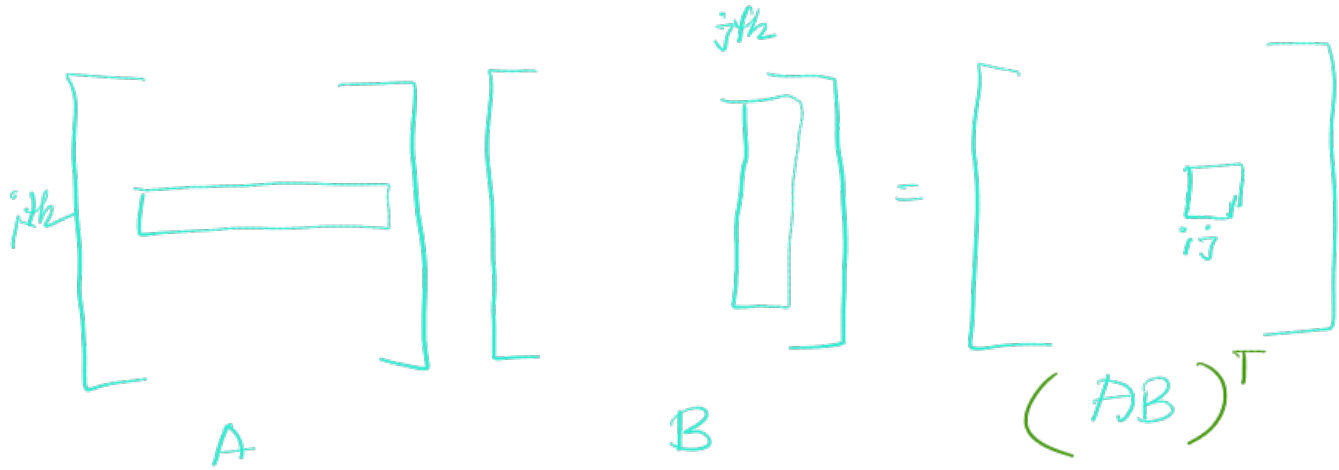
$\begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$ can't be LU decomposed. cz need Row-swap (pivot 0)

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$$

$$P \qquad A \qquad = LDU$$

$$(AB)^T = B^T A^T$$

$m \times n$ $m \times p$



$$AB_{i,j} = row(A, i) \cdot col(B, j)$$

$$(AB)^T_{j,i} = col(B, j) \cdot row(A, i) \quad \text{—— ①}$$

$(B^T A^T)$

$(row(B^T, j) \cdot col(A^T, i)$

↳ col (B, j) · row (A, i) ✓

Vector Spaces → set of vectors

↳ cartesian plane ($\mathbb{R}^2$)

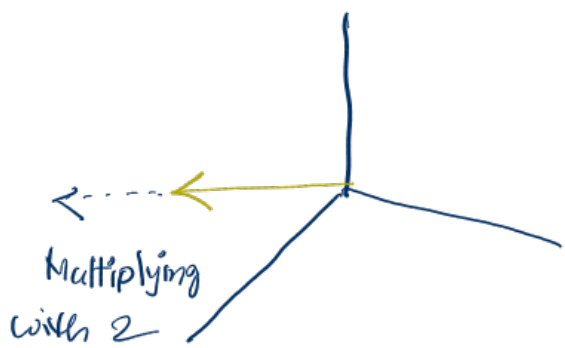
Condition to be a V.S

1) It has to be closed under multiplication. (multiply & result is in the same space)
2) " " " addition.

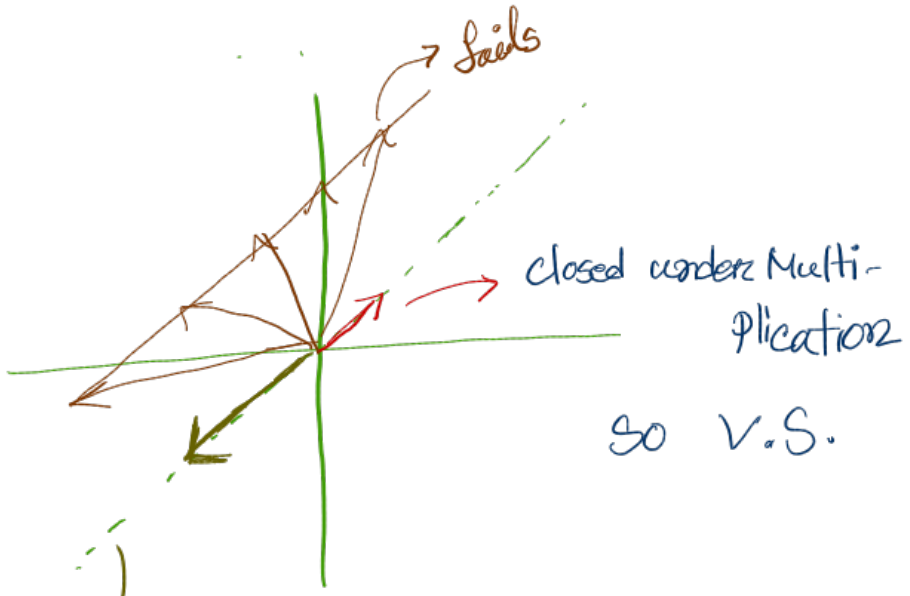
$\mathbb{R}^3$

$\mathbb{R}^n$

$K \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ → set of vectors



↳ it escapes → so not V.S.
not part of the original set.



so V.S.

dotted line → same set: take any vector

from the set to add.

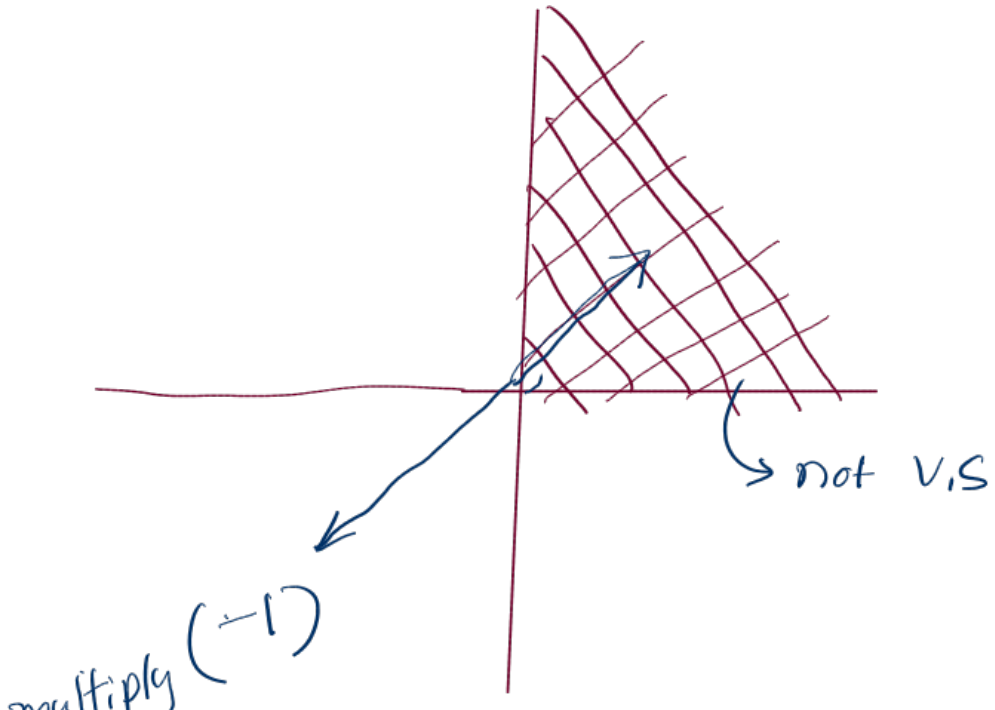
So closed under Addition.

So a vector space.

↳ easy intuition →

must contain origin

otherwise 1st condition will fail.



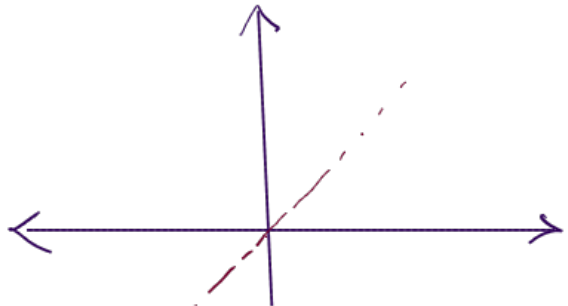
multiply (-1)

and escapes

V.S.

$$\begin{bmatrix} \mathbb{R} \\ \mathbb{R} \end{bmatrix} \rightarrow \text{2D plane}$$

Subspace:



subspace: any line going through origin

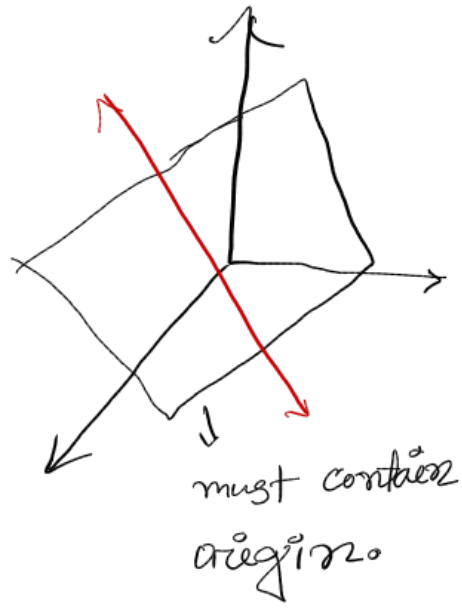
1 space less than the main space

also origin point. (singular subspace)

Spaces in:

$\mathbb{R}^2$ → $\mathbb{R}^2$, $\mathbb{R}^1$, $\mathbb{R}^0$

$\mathbb{R}^3$ → $\mathbb{R}^3$, $\mathbb{R}^2$, $\mathbb{R}^1$, $\mathbb{R}^0$



must contain origin.

Col space & Row space.

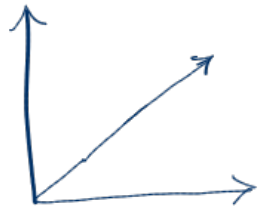
Dimension hopping → separate

$$\begin{bmatrix} 1 & 5 & 6 \\ 2 & 7 & 9 \\ 3 & 8 & 11 \end{bmatrix}$$

2 ind col so 2D plane

A 2D plane in $\mathbb{R}^3$

$C_3 = C_1 + C_2$



Col space.

multiply with 0 → stays in plane

Q]

$$X_1 = \begin{bmatrix} 0 \\ 1 \\ 3 \end{bmatrix} \quad X_2 = \begin{bmatrix} 2 \\ 4 \\ 0 \end{bmatrix}$$

spans space S_1 S_2

1) $S_1 \cap S_2$

2) $[X_1 \quad X_2] \rightarrow \text{col}(A) = S_3$ (col space of A)

A

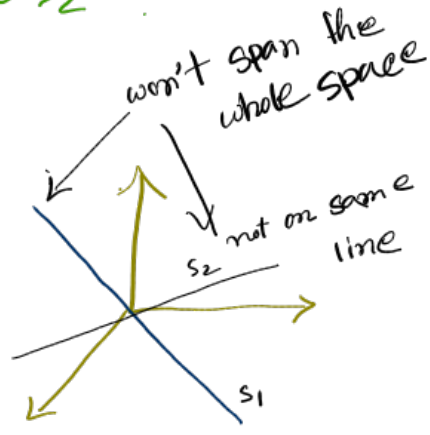
$S_3 = S_1 \cap S_2$ | $S_3 = S_1 \cup S_2$?

S_3 is a plane containing 0 in $\mathbb{R}^3$ | none

1

$$\begin{bmatrix}0\\1\\3\end{bmatrix} \cap \begin{bmatrix}2\\4\\0\end{bmatrix} = \begin{bmatrix}0\\0\\0\end{bmatrix}$$ → 0 vector

line in 3D space.



otherwise $S_1 \cap S_2 = S_1$

2

$$\begin{bmatrix}0 & 2\\1 & 4\\3 & 0\end{bmatrix} = S_1 \cup S_2$$ ✗

3×2

↳ A plane in $\mathbb{R}^3$

↑

S_3

$S_1 \in S_3$

$S_2 \in S_3$

S_1 & S_2 subspace of S_3

NULL space Next class

Col // check.